

Empowering the Underserved, One App at a Timeⁱ

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ABSTRACT

Cellular phones are bringing computing to underserved communities. As people in these communities across the globe lack access to information, but do own cellular phones, apps are filling the gap by delivering meaningful information and important services to improve lives. In this paper, we describe some projects through which students have been helping underserved communities in the United States and globally by developing apps to serve specific social needs.

Categories and Subject Descriptors

K.4 [Computers and Society]: Social Issues, Organizational Impacts

General Terms

Design, Economics, Human Factors

Keywords

Mobile Computing, Mobile Apps, Mobile Computing for Social Benefit, Social Enterprises.

1. INTRODUCTION

Recently, with the advent of smart phones and the widespread usage of apps, cellular phones have become much more than conversation devices, and they are actually changing the way computing serves society. One particular area, in which cellular phones are making a huge difference, is dissemination of information. As people in underserved communities across the globe lack access to information, but do own cellular phones, apps are filling the gap by delivering meaningful information and important services to improve their lives.

Besides providing access to information, phones and SMS communication are helping small businesses to thrive in remote areas, where there are no computers or access to WiFi or the Internet. These small businesses are empowering people in underserved communities, and cellular phones are enabling these businesses to scale by providing them with mechanisms to handle computational-related aspects of a business operation.

Finally, phones are also being used to interface devices, in which case the phone lowers the cost of the device, which also becomes mobile. For example, smart phones are now used to interface medical/diagnostic equipment, such as blood-pressure monitors. Cheaper devices enabled by cellular phones help to improve health in underserved communities.

In this paper we describe some of the projects on which our

group has been working. We divide the projects in three categories: Helping the Homeless, Helping Social Enterprises, and Affordable Devices.

2. Helping the Homeless

For homeless people, cellular phones have a special meaning, as they enable their connection with family and friends. Also, cellular phones constitute a strong tool to help people out of poverty, since it enables them to find jobs and housing. For these reasons, homeless individuals tend to spend a significant portion of their low income on a wireless service.

Research shows that homeless people use cellular phones in a different way than most people, and their needs are very specific [1, 2]. Mostly, they need to talk, text, and browse. They also need the battery to last longer, because they may not have a place to charge the phone every day. We have been working with CTA [1], a local organization that uses technology to help homeless and very-low-income people to understand their needs and develop appropriate applications.

2.1 Homeless Announcement Tool

The announcement tool is a system that enables organizations providing services to homeless individuals to send information about their services to registered people [8]. In particular, informing people of one-time or sporadic services is problematic. For the organizations creating the announcements, we have created a web-based portal allowing them to log in and create their announcement, specifying whether it is related to food, shelter, etc. Once the announcement has been added to the database, an SMS message is sent out to all individuals who have expressed interest in that kind of announcement (food, shelter, etc). This system enables shelters, for example, to announce available beds in a cold night.

2.2 Helping Young Homeless Women

An estimated 2.5 million youth, ages 16-24, experience homelessness in the US each year [3]. These young individuals, particularly women, are considered high-risk for serious health problems, including HIV, STDs, teen pregnancy and teen dating violence. Healthcare providers can play a critical role in providing appropriate care to homeless youth and in linking them to other health services. Research suggests that linking healthcare with other services needed by homeless adolescents, such as shelter or food, may help improve their continuity of care. We have developed a tool formed by two mobile apps that will connect young homeless women to local healthcare providers [9]. One app runs in health clinics to detect high-risk individuals, whom are given the second app, which provides them with info on local health resources. Considering that cellular phones are common among homeless individuals, particularly teenagers, our tool, which is about to be deployed, has the potential to make an impact on these women's lives.

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ACM DEV-5 (2014), December 5–6, 2014, San Jose, CA, USA.

ACM 978-1-4503-2936-1/14/12.

<http://dx.doi.org/10.1145/2674377.2678272>

3. HELPING SOCIAL ENTERPRISES

Social enterprises are businesses whose main goal is to help address problems related to social, economic, environmental, and other issues that developing countries face. We have developed mobile apps and web systems to help social enterprises maintain and scale their business in spite of their lack of access to computers and Wifi or the Internet. We describe some of these apps below.

3.1 mFarm

Farmers in Kenya used to be isolated, with no information about other farmers and their crops and prices. mFarm [4] solved the problem by developing a system through which farmers can use their cellular phone to check information on local crops and prices, enabling them to make better business decisions. As Android phones are becoming widespread in Africa, we have worked with mFarm to develop a more powerful Android interface for their system.

3.2 SOKO

SOKO [5] is a social enterprise that operates in Africa and enables artisans in impoverished communities to sell jewelry on the Internet. Artisans use their phone to upload pictures and prices of the jewelry to the website, which functions as a window for shoppers all over the world. Their system provides underserved people with extra income. Their initial system used a simple app, which we have extended and ported to Android phones.

3.3 Clinicas del Azucar

Diabetes is a serious problem in Mexico, and Clinicas del Azucar [6] is a social enterprise created to treat diabetic people, and an important part of this treatment consists of educating patients. We have developed a system to enable them to make educational articles available to smartphones. The system contains a web interface, on which articles are posted, and an app is used to browse and read these articles, which may be rated and shared.

3.4 Salauno

Salauno [7] is a social enterprise focused in providing inexpensive treatment for cataracts in Mexico. We have developed a tool for them to interact with their patients by sending SMS messages with important treatment reminders.

4. AFFORDABLE DEVICES

Cellular phones are providing certain devices with an inexpensive interface, making them affordable to emerging markets. Health care in underserved communities has benefitted tremendously from this trend, as clinics in remote villages can afford these simplified, mobile versions of expensive diagnostic equipment, such as ultrasound and electrocardiogram devices.

Cellular phones have also been able to substitute certain devices, making them as inexpensive as an app. As both their computational capacity and number of sensors available increase, cellular phones have the potential to substitute a large number of expensive devices. We have been developing apps to enable both smart and feature phones to emulate expensive devices. Two examples are described below.

4.1 HearThat?

Hearing loss is a problem that affects people of all ages all over the world. However, because it is not a life-threatening problem, and resources in emerging markets are scarce, audiometric devices for hearing-loss diagnostics are not typically available in

underserved communities. Considering this scenario, we have implemented an app for hearing-loss pre-screening [10]. Our app can execute in a broad range of phones, which are available everywhere, providing individuals in underserved communities with access to hearing-loss diagnostics.

4.2 Hearing Aid for Everyone

Hearing loss affects people of all ages everywhere. Hearing aid devices have been evolving over the years, and very sleek devices are now available at reasonable prices. However, these devices are still not available for people in underserved communities. We have developed a mobile app that enables people to use their cellular phone as a hearing-aid device. Considering that cellular phones are available everywhere, our app will provide people everywhere with free access to hearing aid. Our app is also useful in emergency situations, when someone who has a hearing-aid device loses it or runs out of battery.

5. CONCLUSION

Cellular phones constitute a powerful tool to not only help underserved people, but also to empower them out of poverty. We have been successfully working with local and international organizations to enable them to use phones for social benefit.

6. ACKNOWLEDGMENTS

We would like to thank our colleagues from Santa Clara University: Radha Basu and Elizabeth Sweeny, from the Frugal Innovation Lab, Keith Warner, from the Center for Science Technology and Society, and Natalie Linnell, from Math/Computer Science. We would also like to thank CTA and YTH, in particular Allan Baez and Deb Levine, for their support. Finally, we would like to acknowledge the students that worked on these projects and helped to deploy these tools [11].

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ⁱ This work has been supported by funds from the following sources: SCU School of Engineering, SCU Frugal Innovation Lab, SCU Center for Science Technology and Society, SCU/CSTS Roelandts Grants, Cisco Systems, and Vodafone Americas Foundation.